

# @sigstore/core has DSSE payloadType type-binding failure

Report Type: Weekly · Generated: June 26, 2026 20:50 UTC · Coverage: Jun 26 - Jun 26, 2026

|                       |                          |                                 |                      |
|-----------------------|--------------------------|---------------------------------|----------------------|
| 1<br>Total Advisories | 0<br>Critical            | 0<br>High                       | 1<br>Medium          |
| MEDIUM<br>Severity    | 5.4<br>CVSS / Risk Score | TLP:CLEAR<br>TLP Classification | PRO<br>Customer Tier |

## Executive Summary

### Impact The `preAuthEncoding` function in `@sigstore/core` uses Node.js `ascii` encoding when converting the PAE (Pre-Authentication Encoding) string to bytes. This allows `payloadType` to be mutated after signing without invalidating the signature, breaking the type-binding guarantee that DSSE is designed to provide. In `packages/core/src/dsse.ts`, the PAE function builds a string containing `payloadType` and then encodes it with `Buffer.from(prefix, 'ascii')`. In Node.js, `ascii` encoding for string-to-Buffer is equivalent to `latin1`, which **truncates characters above U+00FF to their low byte**. This means for any ASCII character, there exist Unicode characters (at U+01xx, U+02xx, etc.) that produce the identical encoded byte:

|  | Original                      | Codepoint | Mutant | Codepoint | Encoded byte |      |  |     |        |   |        |      |  |    |
|--|-------------------------------|-----------|--------|-----------|--------------|------|--|-----|--------|---|--------|------|--|----|
|  | ----- ----- ----- ----- ----- | `t`       | U+0074 | █         | U+0174       | 0x74 |  | `e` | U+0065 | █ | U+0165 | 0x65 |  | An |

**CONFIDENTIALITY NOTICE:** This report is classified PRO tier and intended exclusively for the named recipient. Redistribution, resale, or disclosure to unauthorized parties is strictly prohibited. © 2026 CYBERDUDEBIVASH SENTINEL APEX. All rights reserved.

## Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

| Advisory Title                                   | Severity | CVSS | EPSS | AI Summary                                                                    |
|--------------------------------------------------|----------|------|------|-------------------------------------------------------------------------------|
| @sigstore/core has DSSE payloadType type-binding | MEDIUM   | 5.4  | -    | ### Impact<br>The `preAuthEncoding` function in `@sigstore/core` uses Node.js |

## MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

## Threat Actor Intelligence

Attribution analysis across advisories in this report period.

| Threat Actor   | Count | Associated CVEs / Indicators |
|----------------|-------|------------------------------|
| CDB-UNATTR-CVE | 1     | -                            |

## Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intel/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

## Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

### Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - CDB-ADVISORY Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: Detects exploitation indicators related to CDB-ADVISORY.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/06/26
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CDB-ADVISORY'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium
```

### Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - @sigstore/core has DSSE payloadType type-binding failure
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "@sigstore/core has DSSE payloadType type"
or AlertName has "@sigstore/core has DSSE payloadType type"
or ExtendedProperties has "@sigstore/core has DSSE payloadType type"
| extend Rule = "APEX-ADVISORY", Severity = "MEDIUM"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

## Incident Response Playbook

### Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield @sigstore/core has DSSE payloadType type instances exposed to the internet.
3. Enable verbose access logging on all @sigstore/core has DSSE payloadType type endpoints immediately.
4. Pull last 72h of web server and application logs for forensic baselining.

- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for @sigstore/core has DSSE payloadType type or implement recommended mitigation.
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

|                                                                |                                                             |                                                             |                                                      |
|----------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------|
| <div>\$150K - \$900K</div> <div>Breach Cost Range (FAIR)</div> | <div>\$380K</div> <div>IBM Cost of Breach 2025 Median</div> | <div>\$120K/day</div> <div>Business Interruption Cost</div> | <div>\$85K</div> <div>Regulatory Fine Exposure</div> |
|----------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------|

Medium severity events require internal investigation and targeted customer notification. Average cost includes 340 hours of engineering time.

**Remediation Cost Estimate:** \$55K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

| Framework      | Reference     | Obligation                                                               |
|----------------|---------------|--------------------------------------------------------------------------|
| PCI-DSS v4.0   | Req. 6.3.3    | Non-critical patches applied within 3 months                             |
| NIST CSF 2.0   | ID.RA / PR.PS | Vulnerability assessment and platform security policies                  |
| ISO 27001:2022 | A.8.8         | Vulnerability management programme with documented remediation timelines |

## Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Monitor @sigstore/core has DSSE payloadType type-binding failure for escalation.
2. Apply patches in next maintenance window.
3. Apply vendor patch for @sigstore/core has DSSE payloadType type-binding failure immediately - check NVD for official fix guidance.
4. Enable enhanced logging and alerting on all systems running affected software versions.
5. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
6. Audit all internet-facing instances of affected products and confirm patch deployment.
7. Conduct a threat hunt using MITRE ATT&CK techniques mapped in this advisory.

Appendix - Report Metadata & Legal

| Field          | Value                                         |
|----------------|-----------------------------------------------|
| Report ID      | intel--40b3e93c20c84682b0206200               |
| Report Type    | Weekly                                        |
| Platform       | CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE |
| Generated At   | 2026-06-26T20:50:39.56929                     |
| Customer Tier  | PRO                                           |
| Customer Email | -                                             |
| TLP            | TLP: CLEAR                                    |
| STIX Version   | 2.1                                           |
| ATT&CK Version | v15                                           |

Legal & Usage Notice

This threat intelligence report is produced by CYBERDUDEBIVASH SENTINEL APEX, operated by Bivash Nath (GSTIN: 21ARKPN8270G1ZP). All intelligence is aggregated from public vulnerability databases (NVD, CVSS), threat feeds, and proprietary AI enrichment pipelines. This report is provided AS-IS for informational and defensive security purposes only. CYBERDUDEBIVASH accepts no liability for actions taken based on this intelligence. Redistribution, resale, or disclosure to unauthorized parties is strictly prohibited without written consent. For licensing inquiries: enterprise@cyberdudebivash.com

CYBERDUDEBIVASH® SENTINEL APEX // CONFIDENTIAL - FOR AUTHORIZED USE ONLY